

Journey from OLS to Regularization: The Life of Penalties and Feature Prominence

Algebraic, Statistical, and Geometric Perspectives

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1. Recap: Solving Linear Systems $Ax = b$

Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$. We analyze cases based on b 's relationship with $\text{Col}(A)$:

1. $b \in \text{Col}(A)$: Consistent system
2. $b \notin \text{Col}(A)$: Inconsistent system and hence we approximate the best solution (Least squares)

1.1 $b \in \text{Col}(A)$: Consistent System

Solving using the **Summation form of SVD**

- ▶ In class we solved this by basis($\text{Colsp}(A)$) as eigenvectors of $(A^\top A)$ which in this case are also the columns of U i.e. $Au = \sigma v$
- ▶ Simultaneously we must take basis($\text{Colsp}(A^\top)$) as eigenvector of $(AA^\top)$ which in this case are the columns of V i.e $Av = \sigma u$
- ▶ For a column rank matrix the solutions will be unique and hence we neglected the summation term in x contributed from the basis($\text{Nullsp}(A)$)
- ▶ The value of x concluded was :

$$x_{\text{OLS}} = \sum_{i=1}^r \frac{1}{\sigma_i} v_i \left(u_i^\top b \right)$$

1.2 $b \in \text{Col}(A)$: Consistent System

$\text{rank}(A) = n$ (Full Column Rank) Solving using the **Matrix form of SVD**

- ▶ Columns of A are linearly independent
- ▶ $A^\top A$ is symmetric positive definite ($A^\top A \succ 0$)

Solution: $x = (A^\top A)^{-1} A^\top b$ (unique)

Derivation: Let $A = U\Sigma V^\top$ (SVD).

Columns of U span $\text{Col}(A)$, columns of V span $\mathbb{R}^n$.

Write $b = Uc$ (since $b \in \text{Col}(A)$). Then:

$$Ax = b$$

$$U\Sigma V^\top x = Uc$$

$$\Sigma V^\top x = c \quad (\text{multiply by } U^\top)$$

$$V^\top x = \Sigma^{-1}c$$

$$x = V\Sigma^{-1}c = V\Sigma^{-1}U^\top b = (A^\top A)^{-1}A^\top b$$

1.3: $\text{rank}(A) = r < \min(m, n)$

- ▶ $\text{Null}(A) \neq \{0\}$: Non-trivial null space exists
- ▶ Infinite solutions: $x = x_c + x_n$, where:
 - ▶ $x_c \in \text{Col}(A^\top)$ (row space component)
 - ▶ $x_n \in \text{Null}(A)$ (null space component)

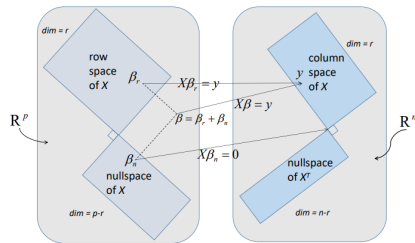
Why infinite? Because

$$A(x_c + x_n) = Ax_c + 0 = b$$

Orthogonal Decomposition Theorem: Any $x \in \mathbb{R}^n$ can be written uniquely as

$$x = x_c + x_n, \quad x_c \perp x_n$$

Interpretation: Data cannot distinguish directions in $\text{Null}(A) \Rightarrow$ identifiability problem.



Infinite solutions along the null space

1.4 $b \notin \text{Col}(A)$: Least Squares Approximation

Minimize squared error: $\min_x \|Ax - b\|_2^2$

Case 1.2.1: $\text{rank}(A) = n$

Project b onto $\text{Col}(A)$: $Pb = A(A^\top A)^{-1}A^\top b$

Orthogonality principle: $b - Ax \perp \text{Col}(A)$

$$A^\top(b - Ax) = 0$$

$$A^\top Ax = A^\top b$$

$$x = (A^\top A)^{-1}A^\top b$$

Geometric: Ax is orthogonal projection of b onto $\text{Col}(A)$.

1.5: $\text{rank}(A) = m$ (More equations than unknowns?)

Actually: if $m > n$ and $\text{rank}(A) = n$, already covered.

If $m < n$ and $\text{rank}(A) = m$ (underdetermined):

Set $x = A^\top y$, then $Ax = AA^\top y = b$ (since $AA^\top$ invertible for full row rank).

Solution: $y = (AA^\top)^{-1}b$, so $x = A^\top(AA^\top)^{-1}b$.

But: This is *one* solution; infinite exist due to null space.

1.6: $\text{rank}(A) = r < \min(m, n)$

Write $x = x_c + x_n$ with $x_c \in \text{Col}(A^\top)$, $x_n \in \text{Null}(A)$.

Error: $\|A(x_c + x_n) - b\|_2^2 = \|Ax_c - b\|_2^2$ (since $Ax_n = 0$).

Thus error independent of x_n ! Minimize by choosing any x_n .

Minimum norm solution: Choose $x_n = 0 \Rightarrow x = x_c$.

Solve $Ax_c = Pb$ where P projects onto $\text{Col}(A)$:

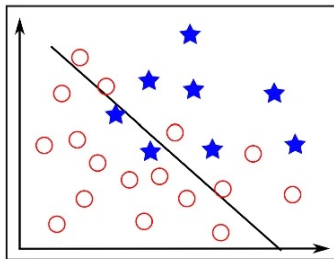
$$x_c = A^+ b \quad (\text{Moore-Penrose pseudoinverse})$$

$$A^+ = V_r \Sigma_r^{-1} U_r^\top \quad \text{using SVD: } A = U_r \Sigma_r V_r^\top$$

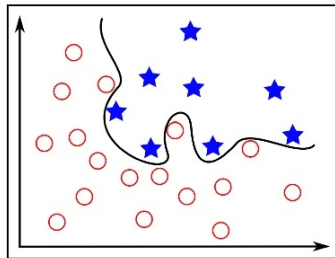
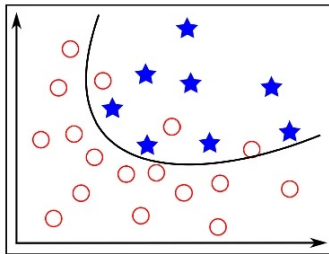
1.7: Can we make this a well-posed problem?

An ill-posed problem is a mathematical or scientific problem that fails one or more conditions for being "well-posed," meaning

- ▶ Lacks a unique solution
- ▶ Have a solution that isn't stable (small data changes cause huge output changes)
 - Matrix Perturbations
- ▶ Computationally infeasible to find (inverse problems where data is indirect or noisy)



Underfitting



Overfitting

2. Why Minimum Norm in Machine Learning?

In ML: $X\beta = y$, but we use notation $Ax = b$.

LIFT Points - **L**ower Bias , **I**mportant features, **F**aster Convergence , **T**raining Stability

1. Low Bias, High Variance Problem:

- ▶ Perfect fit on training data (low bias) but overfits
- ▶ Large coefficients amplify noise
- ▶ Generalization error increases on test data

2. Feature Importance:

- ▶ Regularization shrinks coefficients (some to zero)
- ▶ Automatically reduces weight of collinear features
- ▶ Two approaches: Subset Selection vs Regularization

3. Training Stability:

- ▶ Real data: $A^T A$ often near-singular (ill-conditioned)
- ▶ Causes: collinearity, redundant features, poor scaling
- ▶ Inverse $(A^T A)^{-1}$ explodes $\rightarrow$ numerical instability
- ▶ Small changes in b cause huge changes in x

2.1 Continued: Training Stability & Convergence

1. Faster Gradient Convergence:

- ▶ Condition number $\kappa(A^\top A) = \frac{\sigma_{\max}^2}{\sigma_{\min}^2}$
- ▶ Large $\kappa \rightarrow$ gradient descent oscillates/slow
- ▶ Regularization reduces $\kappa \rightarrow$ faster convergence

There are two reasons why we are often not satisfied with the least squares estimates.

- The first is **prediction accuracy**: the least squares estimates often have low bias but large variance. Prediction accuracy can sometimes be improved by shrinking or setting some coefficients to zero. By doing so we sacrifice a little bit of bias to reduce the variance of the predicted values, and hence may improve the overall prediction accuracy.
- The second reason is **interpretation**. With a large number of predictors, we often would like to determine a smaller subset that exhibit the strongest effects. In order to get the “big picture,” we are willing to sacrifice some of the small details.

2.2 Properties of Minimum Norm Solution

Problem: Among all least-squares solutions, find

$$\min_x \|x\|_2^2 \quad \text{s.t.} \quad \|Ax - b\|_2^2 \text{ is minimized}$$

Key Properties

▶ **Uniqueness:**

- ▶ Error minimization fixes x_c uniquely
- ▶ Minimum norm enforces $x_n = 0$

▶ **Orthogonality:** $x \perp \text{Null}(A)$

▶ **Geometric:** Projection onto $\text{Col}(A^\top)$

Lagrangian:

$$\mathcal{L}(x, \lambda) = \|Ax - b\|_2^2 + \lambda(\|x\|_2^2 - s)$$

Stationarity condition:

$$\nabla_x \mathcal{L} = 2A^\top(Ax - b) + 2\lambda x = 0$$

Lagrange Multiplier Formulation

Equivalent constrained problem:

$$\min_x \|Ax - b\|_2^2 \quad \text{s.t.} \quad \|x\|_2^2 \leq s$$

Solution:

$$(A^\top A + \lambda I)x = A^\top b$$

Insight: Constraint removes null-space freedom $\rightarrow$ selects minimum-energy solution.

Let us understand the problem from another perspective:

We are given a dataset of n observations:

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\},$$

where:

- ▶ $x_i \in \mathbb{R}$ is the input (independent variable),
- ▶ $y_i \in \mathbb{R}$ is the output (dependent variable).

We assume a linear model:

$$\hat{y}_i = mx_i + c,$$

where:

- ▶ m is the slope,
- ▶ c is the intercept,
- ▶ $\hat{y}_i$ is the predicted value.

Our goal is to estimate m and c such that the predictions best fit the observed data.

Why Minimize Squared Error? (Justification of the Loss Function)

Before defining the loss function, we must justify *why* the squared error

$$\ell(e) = e^2$$

is used instead of other possible error measures such as $|e|$ or $|e|^p$. Let the prediction error for the i -th data point be:

$$e_i = y_i - \hat{y}_i.$$

We search for a loss function $\ell(e)$ such that minimizing the total loss

$$\sum_{i=1}^n \ell(e_i)$$

leads to statistically optimal estimates.

Justification : Optimality Under Gaussian Noise (Maximum Likelihood)

Assume

$$y_i = mx_i + c + \varepsilon_i,$$

where ε_i are independent and identically distributed (i.i.d.) random variables with:

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The probability density function of a Gaussian random variable $z \sim \mathcal{N}(\mu, \sigma^2)$ is:

$$p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z - \mu)^2}{2\sigma^2}\right).$$

In our case, the error term $\varepsilon_i = y_i - (mx_i + c)$ has mean 0 and variance σ^2 , so substituting:

$$z = \varepsilon_i, \quad \mu = 0,$$

we obtain:

$$p(\varepsilon_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_i^2}{2\sigma^2}\right).$$

Continuation...

Since $\varepsilon_i = y_i - mx_i - c$, this becomes:

$$p(y_i | x_i, m, c) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - mx_i - c)^2}{2\sigma^2}\right).$$

The conditional probability density of observing y_i given (x_i, m, c) is:

$$p(y_i | x_i, m, c) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - mx_i - c)^2}{2\sigma^2}\right).$$

For the entire dataset, assuming independence of observations, the likelihood function is:

$$\mathcal{L}(m, c) = \prod_{i=1}^n p(y_i | x_i, m, c).$$

Substituting the Gaussian conditional density:

$$\mathcal{L}(m, c) = \prod_{i=1}^n \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - mx_i - c)^2}{2\sigma^2}\right) \right].$$

Using properties of products and exponents:

$$\mathcal{L}(m, c) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2 \right).$$

Taking the natural logarithm of the likelihood:

$$\log \mathcal{L}(m, c) = \log \left(\left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \right) + \log \left(\exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2 \right) \right).$$

Simplifying each term:

$$\log \mathcal{L}(m, c) = n \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2.$$

Since:

$$\log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) = -\frac{1}{2} \log(2\pi\sigma^2),$$

we obtain:

$$\log \mathcal{L}(m, c) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - mx_i - c)^2.$$

Loss Function: Least Squares Criterion

Maximizing $\log \mathcal{L}(m, c)$ is equivalent to minimizing:

$$\sum_{i=1}^n (y_i - mx_i - c)^2.$$

Conclusion which we got now is : Minimizing squared error is exactly equivalent to performing maximum likelihood estimation (MLE), which is statistically optimal.

Define the residual (error) for each data point:

$$e_i = y_i - \hat{y}_i = y_i - (mx_i + c).$$

We define the total squared error (objective function) as:

$$J(m, c) = \sum_{i=1}^n (y_i - mx_i - c)^2.$$

The Ordinary Least Squares (OLS) problem is:

$$\min_{m, c} J(m, c).$$

Partial Derivative with Respect to m

To minimize $J(m, c)$, we compute the partial derivatives with respect to m and c and set them equal to zero.

$$\frac{\partial J}{\partial m} = \frac{\partial}{\partial m} \sum_{i=1}^n (y_i - mx_i - c)^2 = \sum_{i=1}^n 2(y_i - mx_i - c)(-x_i).$$

Setting this to zero:

$$\sum_{i=1}^n x_i(y_i - mx_i - c) = 0.$$

Expanding:

$$\begin{aligned} \sum_{i=1}^n x_i y_i - m \sum_{i=1}^n x_i^2 - c \sum_{i=1}^n x_i &= 0. \\ \Rightarrow m \sum_{i=1}^n x_i^2 + c \sum_{i=1}^n x_i &= \sum_{i=1}^n x_i y_i. \end{aligned} \tag{1}$$

Partial Derivative with Respect to c

$$\frac{\partial J}{\partial c} = \sum_{i=1}^n 2(y_i - mx_i - c)(-1).$$

Setting this to zero:

$$\sum_{i=1}^n (y_i - mx_i - c) = 0.$$

Expanding:

$$\begin{aligned} \sum_{i=1}^n y_i - m \sum_{i=1}^n x_i - nc &= 0. \\ \Rightarrow m \sum_{i=1}^n x_i + nc &= \sum_{i=1}^n y_i. \end{aligned} \tag{2}$$

Solving the System

We now solve equations (1) and (2) simultaneously.

Thus:

$$m = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}.$$

Then:

$$c = \bar{y} - m\bar{x},$$

where:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

Matrix Formulation of Linear Regression

We now generalize to vector/matrix form and re-derive OLS using linear algebra.
Define:

$$\mathbf{X} = \begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix} \in \mathbb{R}^{n \times 2}, \quad \mathbf{w} = \begin{bmatrix} m \\ c \end{bmatrix} \in \mathbb{R}^{2 \times 1}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^{n \times 1}.$$

Then predictions:

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}.$$

Residual vector:

$$\mathbf{r} = \mathbf{y} - \mathbf{X}\mathbf{w}.$$

Objective Function in Matrix Form

The squared error over all samples is:

$$J(\mathbf{w}) = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

The prediction vector is:

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}.$$

Hence,

$$J(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2.$$

By definition of the Euclidean L2 norm for a vector $\mathbf{z}$:

$$\|\mathbf{z}\|_2^2 = \mathbf{z}^\top \mathbf{z}.$$

Applying this definition:

$$\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 = (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}).$$

Gradient and Normal Equations

Now expand the transpose:

$$(\mathbf{y} - \mathbf{X}\mathbf{w})^\top = \mathbf{y}^\top - (\mathbf{X}\mathbf{w})^\top = \mathbf{y}^\top - \mathbf{w}^\top \mathbf{X}^\top,$$

using the identity $(\mathbf{A}\mathbf{b})^\top = \mathbf{b}^\top \mathbf{A}^\top$.

Substituting:

$$J(\mathbf{w}) = (\mathbf{y}^\top - \mathbf{w}^\top \mathbf{X}^\top)(\mathbf{y} - \mathbf{X}\mathbf{w}).$$

Now multiply term-by-term:

$$J(\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}.$$

Since $\mathbf{y}^\top \mathbf{X}\mathbf{w}$ is a scalar, it equals its transpose:

$$\mathbf{y}^\top \mathbf{X}\mathbf{w} = (\mathbf{y}^\top \mathbf{X}\mathbf{w})^\top = \mathbf{w}^\top \mathbf{X}^\top \mathbf{y}.$$

Therefore, the two middle terms are equal and combine:

$$J(\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{w} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}.$$

Closed-Form Solution

Take gradient w.r.t. $\mathbf{w}$:

$$\nabla_{\mathbf{w}} J = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \mathbf{w}.$$

Set to zero:

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y}.$$

These are the **Normal Equations**. If $\mathbf{X}^T \mathbf{X}$ is invertible:

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

This yields:

$$\mathbf{w}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

which gives both m and c .

Assumptions Behind OLS

OLS assumes:

1. Linearity: $y = \mathbf{X}\mathbf{w} + \varepsilon$,
2. Full rank: $\mathbf{X}^\top \mathbf{X}$ is invertible.

Violations of these assumptions lead to instability or non-uniqueness of solutions which motivates regularization (next part).

A Concrete Failure Case of Ordinary Least Squares

Consider a small dataset where the input features are highly correlated:

Index	x_1	x_2	y
1	1.0	1.1	2.1
2	2.0	2.1	4.1
3	3.0	3.1	6.1
4	4.0	4.1	8.1

Observe that:

$$x_2 \approx x_1 + 0.1,$$

so the features are nearly linearly dependent (high multicollinearity).

Let the model be:

$$y = w_1 x_1 + w_2 x_2.$$

Formulation of L2 Regularization

The design matrix is:

$$\mathbf{X} = \begin{bmatrix} 1.0 & 1.1 \\ 2.0 & 2.1 \\ 3.0 & 3.1 \\ 4.0 & 4.1 \end{bmatrix}.$$

Then:

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 30 & 31 \\ 31 & 32.04 \end{bmatrix}.$$

The determinant is:

$$\det(\mathbf{X}^\top \mathbf{X}) = 30 \cdot 32.04 - 31^2 = 961.2 - 961 = 0.2,$$

which is very close to zero.

Implication: $\mathbf{X}^\top \mathbf{X}$ is nearly singular, so:

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

has extremely large entries, leading to unstable, high-variance weight estimates.

Algebraic Derivation of Ridge Regression (Scalar Form)

OLS may produce very large coefficients w_k to interpolate noise.

By penalizing large coefficients, we force smoother, more stable functions — this is the role of L2 regularization. We modify the OLS objective by adding a penalty on the magnitude of the weights:

$$J_{\text{ridge}}(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2,$$

where:

- ▶ $\|\mathbf{w}\|_2^2 = \mathbf{w}^\top \mathbf{w}$,
- ▶ $\lambda \geq 0$ is the regularization hyperparameter.

Consider the simple linear regression model:

$$y_i = mx_i + c + \varepsilon_i.$$

The ridge objective is:

$$J(m, c) = \sum_{i=1}^n (y_i - mx_i - c)^2 + \lambda m^2.$$

Partial Derivative with Respect to m

$$\frac{\partial J}{\partial m} = \sum_{i=1}^n 2(y_i - mx_i - c)(-x_i) + 2\lambda m.$$

Setting to zero:

$$\sum_{i=1}^n x_i(y_i - mx_i - c) = \lambda m.$$

Expanding:

$$\begin{aligned} \sum x_i y_i - m \sum x_i^2 - c \sum x_i &= \lambda m, \\ \Rightarrow m(\sum x_i^2 + \lambda) + c \sum x_i &= \sum x_i y_i. \end{aligned} \tag{1}$$

Partial Derivative with Respect to c

$$\frac{\partial J}{\partial c} = \sum_{i=1}^n 2(y_i - mx_i - c)(-1).$$

Setting to zero:

$$\begin{aligned}\sum y_i - m \sum x_i - nc &= 0, \\ \Rightarrow m \sum x_i + nc &= \sum y_i.\end{aligned}\tag{2}$$

Solving the System

Thus:

$$m_{\text{ridge}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2 + \lambda}$$

and then

$$m_{\text{ridge}} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2 + n\lambda}.$$

Compare this with OLS:

$$m_{\text{OLS}} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}.$$

Observation: Ridge regression inflates the denominator, preventing division by small values — thereby stabilizing the solution.

Matrix-Level Derivation of Ridge Regression

We start with the ridge regression objective:

$$J(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2.$$

Using the identity $\|z\|_2^2 = z^\top z$, we expand both terms:

$$J(\mathbf{w}) = (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) + \lambda \mathbf{w}^\top \mathbf{w}.$$

Expand the quadratic term:

$$(\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}.$$

Since $\mathbf{y}^\top \mathbf{X}\mathbf{w}$ is a scalar, we have:

$$\mathbf{y}^\top \mathbf{X}\mathbf{w} = \mathbf{w}^\top \mathbf{X}^\top \mathbf{y}.$$

Thus the objective becomes:

$$J(\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w} + \lambda \mathbf{w}^\top \mathbf{w}.$$

Lets find $\mathbf{w}$

Compute the gradient with respect to $\mathbf{w}$. Using the identities:

$$\nabla_{\mathbf{w}}(\mathbf{w}^{\top} \mathbf{A} \mathbf{w}) = (\mathbf{A} + \mathbf{A}^{\top}) \mathbf{w}, \quad \nabla_{\mathbf{w}}(\mathbf{b}^{\top} \mathbf{w}) = \mathbf{b},$$

and noting that $\mathbf{X}^{\top} \mathbf{X}$ is symmetric, we get:

$$\nabla_{\mathbf{w}} J = -2\mathbf{X}^{\top} \mathbf{y} + 2\mathbf{X}^{\top} \mathbf{X} \mathbf{w} + 2\lambda \mathbf{w}.$$

Setting the gradient to zero:

$$-2\mathbf{X}^{\top} \mathbf{y} + 2\mathbf{X}^{\top} \mathbf{X} \mathbf{w} + 2\lambda \mathbf{w} = 0.$$

Divide both sides by 2:

$$\mathbf{X}^{\top} \mathbf{X} \mathbf{w} + \lambda \mathbf{w} = \mathbf{X}^{\top} \mathbf{y}.$$

Factor out $\mathbf{w}$:

$$(\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I}) \mathbf{w} = \mathbf{X}^{\top} \mathbf{y}.$$

Provided $\lambda > 0$, the matrix $(\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I})$ is always invertible. Therefore:

$$\mathbf{w}_{\text{ridge}} = (\mathbf{X}^{\top} \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^{\top} \mathbf{y}$$

When Is L2 Regularization Required?

L2 regularization is particularly necessary when:

1. $\mathbf{X}^\top \mathbf{X}$ is singular or nearly singular,
2. There is multicollinearity among features,
3. The number of features d is comparable to or larger than n ,
4. The model exhibits overfitting,
5. Numerical stability is required.
6.
 - ▶ $\lambda = 0$ reduces to OLS.
 - ▶ Interpretation of the Hyperparameter
 $\lambda \rightarrow \infty$ forces $\mathbf{w} \rightarrow \mathbf{0}$.
 - ▶ Larger λ :
 - ▶ Increases bias,
 - ▶ Decreases variance,
 - ▶ Improves generalization in noisy or ill-conditioned settings.

Thus, λ controls the **bias–variance tradeoff**.

2.3 Ridge Regression: 2D vs Multi-D View

2D (Single Feature)

$$A = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad x = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

OLS:

$$m_{\text{OLS}} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

Ridge:

$$m_{\text{ridge}} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2 + n\lambda}$$

Effect: denominator inflation $\Rightarrow$ slope shrinkage

Multi-D (General Case)

$$\min_x \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

Normal Eq.:

$$(A^\top A + \lambda I)x = A^\top b$$

Solution:

$$x_{\text{ridge}} = (A^\top A + \lambda I)^{-1} A^\top b$$

Meaning: eigenvalues lifted, null directions suppressed, unique solution

2.3 Effect on coefficients in case of multi dimension

Let $A = U\Sigma V^\top$ (full SVD), $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$, $\Sigma \in \mathbb{R}^{m \times n}$

OLS Solution: $x_{\text{OLS}} = (A^\top A)^{-1} A^\top b = V \Sigma^{-1} U^\top b$

where $\Sigma^{-1} = \text{diag}(1/\sigma_1, \dots, 1/\sigma_r, 0, \dots, 0)$

Ridge Solution: $x_{\text{ridge}} = (A^\top A + \lambda I)^{-1} A^\top b = V(\Sigma^\top \Sigma + \lambda I)^{-1} \Sigma^\top U^\top b$

Component-wise:

$$x_{\text{ols}} = \sum_{i=1}^r \frac{1}{\sigma_i} v_i (u_i^\top b)$$

$$x_{\text{ridge}} = \sum_{i=1}^r \left(\frac{\sigma_i}{\sigma_i^2 + \lambda} \right) v_i (u_i^\top b)$$

Compare: OLS uses $1/\sigma_i$, Ridge uses $\sigma_i/(\sigma_i^2 + \lambda)$

3.1 General Tikhonov Regularization

General Form: Balance data fidelity and solution structure:

$$\min_x \|Ax - b\|_2^2 + \|Tx\|_2^2$$

Closed-Form Solution:

$$x = (A^\top A + T^\top T)^{-1} A^\top b$$

Interpretation:

- ▶ $\|Ax - b\|_2^2$: fit to observed data
- ▶ $\|Tx\|_2^2$: encodes prior belief on solution
- ▶ T : selects which directions in x are penalized

3.2 Choosing the Regularization Matrix T

Standard Ridge: $T = \sqrt{\lambda}I$

- ▶ Uniform penalty on all coefficients
- ▶ Controls overall energy of x
- ▶ Bias–variance tradeoff without feature preference

Weighted Ridge: $T = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$

- ▶ Different penalty per feature
- ▶ Large $\lambda_i \Rightarrow$ *feature is noisy or unreliable*
- ▶ Common in econometrics

Derivative-Based Regularization: $T = \sqrt{\lambda}L$

- ▶ $L = I$: penalizes magnitude (zero-order)
- ▶ $L = [1 \ -1]$: penalizes jumps (first-order smoothness)
- ▶ $L = [1 \ -2 \ 1]$: penalizes curvature (second-order smoothness)

3.3 Applications of Tikhonov Regularization

Image Deblurring

Model: $Ax = b$, where A blurs the image.

- ▶ Inversion amplifies high-frequency noise
- ▶ Choosing $T = L$ suppresses pixel-scale oscillations
- ▶ Result: stable and visually smooth reconstruction

Signal Reconstruction

Solve noisy inverse problems using:

$$\min_x \|Ax - b\|_2^2 + \lambda \|Lx\|_2^2$$

- ▶ First-order L : piecewise smooth signals
- ▶ Second-order L : smooth curvature (splines)

Geophysical & Medical Imaging

- ▶ Problems are severely ill-posed
- ▶ Many solutions fit the same measurements

Motivation for L1 Regularization

In the previous section, we studied Ridge Regression (L2 regularization), which improves numerical stability and reduces overfitting by shrinking the weights. However, Ridge regression typically does not set weights exactly to zero. In many real-world datasets, especially when the number of features is large, we may want a model that automatically selects only the most important features.

This motivates **L1 regularization**, also known as **Lasso Regression**, which encourages **sparsity** in the model parameters (i.e., some weights become exactly zero).

Formulation of Lasso Regression

Let the linear regression model be:

$$\hat{y} = Xw$$

where:

$$X \in \mathbb{R}^{n \times d}, \quad w \in \mathbb{R}^{d \times 1}, \quad y \in \mathbb{R}^{n \times 1}.$$

The Lasso objective function is defined as:

$$J_{\text{lasso}}(w) = \|y - Xw\|_2^2 + \lambda \|w\|_1$$

where:

$$\|w\|_1 = \sum_{j=1}^d |w_j|$$

and $\lambda \geq 0$ is the regularization hyperparameter.

Key Properties of Lasso

- ▶ When $\lambda = 0$, Lasso reduces to Ordinary Least Squares (OLS).
- ▶ As λ increases, more weights are forced toward zero.
- ▶ Lasso can produce a **sparse model**, meaning some $w_j = 0$ exactly.
- ▶ This sparsity provides **feature selection**, which can improve interpretability.
- ▶ Lasso may behave unstably when features are **highly correlated** (may select one and drop others).

Comparison: Ridge (L2) vs Lasso (L1)

► Ridge (L2):

$$J_{\text{ridge}}(w) = \|y - Xw\|_2^2 + \lambda \|w\|_2^2$$

Ridge shrinks weights smoothly but generally does not make them exactly zero. This is because the ℓ_2 penalty is differentiable:

$$\frac{d}{dw_j}(w_j^2) = 2w_j,$$

so coefficients are continuously pushed toward 0 but rarely become exactly 0.

► Lasso (L1):

$$J_{\text{lasso}}(w) = \|y - Xw\|_2^2 + \lambda \|w\|_1$$

Lasso shrinks weights and can force some coefficients to become exactly zero. The ℓ_1 penalty has a non-differentiable kink at $w_j = 0$:

$$\|w\|_1 = \sum_{j=1}^d |w_j|, \quad \frac{d}{dw_j}|w_j| = \begin{cases} +1 & w_j > 0 \\ -1 & w_j < 0 \end{cases}$$

which allows coefficients to be driven exactly to 0, producing sparsity.

Why Lasso Does Not Have a Closed-Form Solution

For Ridge regression, the penalty term $\|w\|_2^2 = w^T w$ is differentiable, which allows a closed-form solution:

$$w_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y.$$

However, for Lasso regression, the penalty term $\|w\|_1 = \sum |w_j|$ contains an absolute value function, which is **not differentiable at** $w_j = 0$. Therefore, Lasso generally does not have a closed-form solution and must be solved using iterative optimization methods.

Interpretation of the Hyperparameter λ

The hyperparameter $\lambda \geq 0$ controls the trade-off between **data fitting** and **model sparsity**:

$$J_{\text{lasso}}(w) = \|y - Xw\|_2^2 + \lambda \|w\|_1.$$

- ▶ $\lambda = 0$: No regularization, equivalent to OLS.
- ▶ **Moderate** λ : Encourages sparsity by shrinking weak coefficients toward 0, often improving generalization and interpretability.
- ▶ **Very large** λ : Dominates the objective, forcing many $w_j \approx 0$ and leading to underfitting (high bias).

When Lasso is (and is not) Ideal

- ▶ Lasso is useful when only a few features are truly important (sparse signal). However, if the true signal is **dense** (many small but non-zero coefficients),
- ▶ Lasso may set useful coefficients to zero, leading to underfitting. In such cases, Ridge regression may perform better.
- ▶ Connection to Matrix Perturbation and Stability In real-world datasets, the observed data is often noisy, meaning the feature matrix and labels may be perturbed:

$$X \rightarrow X + \Delta X, \quad y \rightarrow y + \Delta y.$$

- ▶ If X is ill-conditioned (e.g., due to multicollinearity), small perturbations can cause large changes in the estimated coefficients w . Regularization reduces this sensitivity by controlling model complexity:
- ▶ Ridge improves numerical stability via the ℓ_2 penalty, while Lasso promotes sparse and more robust solutions via the ℓ_1 penalty.

Linear Regression Model

Given a dataset $\{(x_i, y_i)\}_{i=1}^n$ with p predictors:

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip}), \quad y_i \in \mathbb{R}$$

The linear model is: $\hat{y}_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j$ Define the residual: $\varepsilon_i = y_i - \hat{y}_i$

The Ordinary Least Squares objective is:

$$\min_{\beta_0, \beta} \frac{1}{2n} \sum_{i=1}^n \varepsilon_i^2 = \frac{1}{2n} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j \right)^2$$

Why Square the Residuals?

- ▶ Ensures non-negative loss
- ▶ Penalizes large deviations heavily
- ▶ Leads to convex optimization

Expansion of RSS Term

Expand the square:

$$(y_i - \hat{y}_i)^2 = y_i^2 - 2y_i \sum_{j=1}^p x_{ij}\beta_j + \left(\sum_{j=1}^p x_{ij}\beta_j \right)^2$$

Summing over all observations:

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum y_i^2 - 2 \sum y_i \mathbf{x}_i^T \boldsymbol{\beta} + \sum \boldsymbol{\beta}^T \mathbf{x}_i \mathbf{x}_i^T \boldsymbol{\beta}$$

This quadratic structure explains why OLS has a closed-form solution.

L2 Regularization (Ridge Regression)

Objective Function:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \frac{\lambda}{2} \sum_{j=1}^p \beta_j^2 \right\}$$

Meaning of L2 Penalty:

$$\|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2$$

- ▶ Penalizes large coefficients
- ▶ Encourages smooth solutions
- ▶ Ensures strict convexity

Gradient of L2 Term:

$$\frac{\partial}{\partial \beta_j} \left(\frac{\lambda}{2} \beta_j^2 \right) = \lambda \beta_j$$

L1 Regularization (Lasso Regression)

Objective Function:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Nature of Absolute Value:

$$|\beta_j| = \begin{cases} \beta_j, & \beta_j > 0 \\ -\beta_j, & \beta_j < 0 \end{cases}$$

Subgradient of L1:

$$\partial|\beta_j| = \begin{cases} \{1\}, & \beta_j > 0 \\ [-1, 1], & \beta_j = 0 \\ \{-1\}, & \beta_j < 0 \end{cases}$$

This non-differentiability at zero creates sparsity.

Elastic Net Regression

Full Algebraic Objective:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - x_i^T \beta)^2 + \lambda \left(\alpha \sum_{j=1}^p |\beta_j| + \frac{1-\alpha}{2} \sum_{j=1}^p \beta_j^2 \right) \right\}$$

- ▶ n : number of training samples
- ▶ p : number of features
- ▶ $y_i \in \mathbb{R}$: response variable for i -th observation
- ▶ $x_i \in \mathbb{R}^p$: feature vector of i -th observation
- ▶ $\beta = (\beta_1, \beta_2, \dots, \beta_p)^T$: coefficient vector
- ▶ β_j : weight for j -th feature
- ▶ $\lambda \geq 0$: regularization parameter
- ▶ $\alpha \in [0, 1]$: mixing parameter

Elastic Net Regression (Continued)

- ▶ $\alpha = 1$: Lasso regression (pure L1 penalty)
- ▶ $\alpha = 0$: Ridge regression (pure L2 penalty)
- ▶ $0 < \alpha < 1$: Elastic Net regression

Penalty Decomposition:

$$\text{Penalty} = \underbrace{\lambda \alpha \|\beta\|_1}_{\text{sparsity}} + \underbrace{\lambda \frac{1 - \alpha}{2} \|\beta\|_2^2}_{\text{stability}}$$

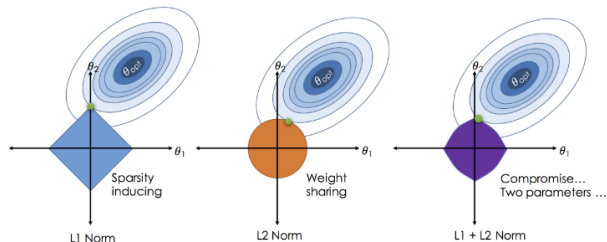
Let:

$$X \in \mathbb{R}^{n \times p}, \quad y \in \mathbb{R}^n, \quad \beta \in \mathbb{R}^p$$

Elastic Net objective in matrix form:

$$\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right) \right\}$$

Geometric Comparison: Ridge vs Lasso vs OLS



Comparison of Ridge, Lasso, Elastic Net, OLS

Key Insight:

- ▶ OLS: Minimizes error only (unconstrained)
- ▶ Ridge: Error + ℓ_2 penalty
→ *smoothshrinkage*
- ▶ Lasso: Error + ℓ_1 penalty
→ *sparsesolutions*

Coordinate Descent Derivation

Elastic Net optimization is difficult to solve jointly for all coefficients because the L1 penalty is not differentiable at zero. Coordinate descent addresses this by optimizing one coefficient at a time while keeping all others fixed.

Why rewrite the objective in terms of a partial residual?

The full response y_i depends on all regression coefficients through the linear model

$$y_i \approx \sum_{k=1}^p x_{ik} \beta_k.$$

When updating a single coefficient β_j using coordinate descent, all other coefficients $\{\beta_k\}_{k \neq j}$ are held fixed. Their contribution to the prediction can therefore be treated as known and subtracted from the response.

Partial Residual

This leads to the definition of the *partial residual*, which isolates the effect of β_j :

$$r_i^{(j)} = y_i - \sum_{k \neq j} x_{ik} \beta_k.$$

- Thus, $r_i^{(j)}$ represents the portion of y_i that remains unexplained after accounting for all predictors except the j -th one. - With all other coefficients fixed, the prediction depends only on β_j through $x_{ij}\beta_j$. The Elastic Net objective function restricted to β_j becomes:

$$\mathcal{L}_j(\beta_j) = \frac{1}{2n} \sum_{i=1}^n \left(r_i^{(j)} - x_{ij}\beta_j \right)^2 + \lambda \alpha |\beta_j| + \lambda \frac{1 - \alpha}{2} \beta_j^2$$

This reduces the original multivariate optimization problem to a one-dimensional problem in β_j .

Expansion of the Squared Error Term

Expanding the squared error term gives:

$$\left(r_i^{(j)} - x_{ij}\beta_j\right)^2 = \left(r_i^{(j)}\right)^2 - 2x_{ij}r_i^{(j)}\beta_j + x_{ij}^2\beta_j^2$$

Substituting this expansion into the objective function yields:

$$\mathcal{L}_j(\beta_j) = \frac{1}{2n} \sum_{i=1}^n \left(r_i^{(j)}\right)^2 - \frac{1}{n} \sum_{i=1}^n x_{ij}r_i^{(j)}\beta_j + \frac{1}{2n} \sum_{i=1}^n x_{ij}^2\beta_j^2 + \lambda\alpha|\beta_j| + \lambda\frac{1-\alpha}{2}\beta_j^2$$

Dropping Constant Terms

The term

$$\frac{1}{2n} \sum_{i=1}^n \left(r_i^{(j)} \right)^2$$

does not involve β_j and is therefore constant with respect to β_j .

Since constants do not affect the location of the minimum of an optimization problem, this term can be safely dropped without changing the optimal value of β_j .

After removing constant terms and combining quadratic terms, the objective simplifies to:

$$\mathcal{L}_j(\beta_j) = -\rho_j \beta_j + \frac{A_j}{2} \beta_j^2 + \lambda \alpha |\beta_j|$$

where:

$$\rho_j = \frac{1}{n} \sum_{i=1}^n x_{ij} r_i^{(j)}, \quad A_j = \frac{1}{n} \sum_{i=1}^n x_{ij}^2 + \lambda(1 - \alpha)$$

Soft-Thresholding Solution (Step-by-Step Derivation)

After fixing all coefficients except β_j , the optimization problem reduces to minimizing a one-dimensional convex function:

$$\min_{\beta_j} \left(\frac{A_j}{2} \beta_j^2 - \rho_j \beta_j + \lambda \alpha |\beta_j| \right)$$

where:

$$\rho_j = \frac{1}{n} \sum_{i=1}^n x_{ij} r_i^{(j)}, \quad A_j = \frac{1}{n} \sum_{i=1}^n x_{ij}^2 + \lambda(1 - \alpha)$$

The presence of the absolute value term $|\beta_j|$ makes the function non-differentiable at $\beta_j = 0$. To solve this, we analyze the problem by considering three cases.

Case 1: $\beta_j > 0$

If $\beta_j > 0$, then $|\beta_j| = \beta_j$. The objective function becomes:

$$f(\beta_j) = \frac{A_j}{2} \beta_j^2 - \rho_j \beta_j + \lambda \alpha \beta_j$$

Taking the derivative and setting it to zero:

$$\frac{df}{d\beta_j} = A_j \beta_j - \rho_j + \lambda \alpha = 0$$

Solving for β_j :

$$\beta_j = \frac{\rho_j - \lambda \alpha}{A_j}$$

This solution is valid only if $\beta_j > 0$, which requires:

$$\rho_j > \lambda \alpha$$

Case 2: $\beta_j < 0$

If $\beta_j < 0$, then $|\beta_j| = -\beta_j$. The objective function becomes:

$$f(\beta_j) = \frac{A_j}{2}\beta_j^2 - \rho_j\beta_j - \lambda\alpha\beta_j$$

Taking the derivative:

$$\frac{df}{d\beta_j} = A_j\beta_j - \rho_j - \lambda\alpha = 0$$

Solving:

$$\beta_j = \frac{\rho_j + \lambda\alpha}{A_j}$$

This solution is valid only if $\beta_j < 0$, which requires:

$$\rho_j < -\lambda\alpha$$

Case 3: $\beta_j = 0$

If $-\lambda\alpha \leq \rho_j \leq \lambda\alpha$, the solutions from Case 1 and Case 2 violate their sign assumptions. Therefore, the minimum occurs at:

$$\beta_j = 0$$

Final Soft-Thresholding Solution Combining all three cases, the coordinate update rule can be written compactly as:

$$\beta_j = \frac{\text{sign}(\rho_j) \max(|\rho_j| - \lambda\alpha, 0)}{A_j}$$

This operator is known as the **soft-thresholding operator**. It shows that coefficients whose correlation with the residual is smaller than the threshold $\lambda\alpha$ are set exactly to zero, resulting in automatic feature selection.

The Population vs. Sample Regression Model

► Population Model (True Relationship):

$$y = \beta_0 + \beta_1 x + \varepsilon, \quad \varepsilon \sim (0, \sigma^2 I)$$

where β_0, β_1 are unknown true parameters, and errors ε are i.i.d. with:

$$E(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0$$

► Mean and Variance of y given x : $E(y|x) = \beta_0 + \beta_1 x$, $\text{Var}(y|x) = \sigma^2$

► Sample Model (Estimated from Data): $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ Goal: Find estimators $\hat{\beta}_0, \hat{\beta}_1$ that are “close” to β_0, β_1 .

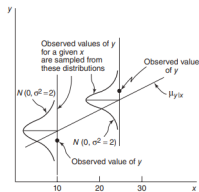


Figure 1.2 How observations are generated in linear regression.

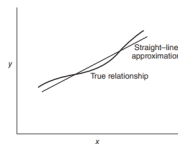


Figure 1.3 Linear regression approximation of a complex relationship.

Ordinary Least Squares (OLS): Best Linear Unbiased Estimator (BLUE)

▶ OLS estimator: $\hat{\beta}_{OLS} = (X'X)^{-1}X'y$.

▶ **Unbiasedness:**

$$\begin{aligned}E(\hat{\beta}_{OLS}) &= E[(X'X)^{-1}X'(X\beta + \varepsilon)] \\&= \beta + (X'X)^{-1}X'E(\varepsilon) = \beta\end{aligned}$$

▶ **Variance:**

$$\begin{aligned}\text{Var}(\hat{\beta}_{OLS}) &= \text{Var}[(X'X)^{-1}X'y] \\&= (X'X)^{-1}X' \text{Var}(y) X(X'X)^{-1} \\&= \sigma^2(X'X)^{-1}\end{aligned}$$

▶ **Gauss-Markov Theorem:** Among all linear unbiased estimators, OLS has the **minimum variance** $\Rightarrow$ it is **BLUE**.

▶ **Dilemma:** Low bias, but variance $\sigma^2(X'X)^{-1}$ can be **large** if X is ill-conditioned (multicollinearity). High variance leads to poor prediction on new data.

Ridge Regression: Bias-Variance Trade-off

- ▶ Ridge estimator: $\hat{\beta}_{Ridge} = (X'X + \lambda I)^{-1}X'y$, $\lambda \geq 0$.

- ▶ **Bias:**

$$\begin{aligned} E(\hat{\beta}_{Ridge}) &= (X'X + \lambda I)^{-1}X'X\beta \\ &\neq \beta \quad (\text{Biased estimator for } \lambda > 0) \end{aligned}$$

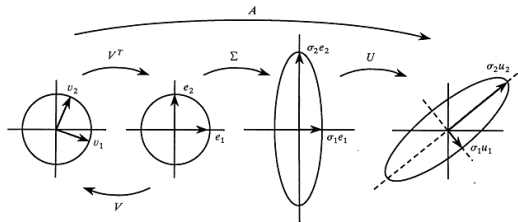
- ▶ **Variance:**

$$Var(\hat{\beta}_{Ridge}) = \sigma^2(X'X + \lambda I)^{-1}X'X(X'X + \lambda I)^{-1}$$

- ▶ **Key Insight** (The “L” Point: **L**ow variance through **L**2 penalty):

- ▶ Adding λI improves conditioning of $(X'X + \lambda I)$.
- ▶ **Shrinks coefficients** towards zero, introducing **small bias**.
- ▶ Drastically **reduces variance** compared to OLS.

Singular Values: Geometry Behind Regression



Let $A \in \mathbb{R}^{m \times n}$ and let its full SVD be : $A = U\Sigma V^T$

Geometric meaning:

- ▶ V^T : rotates parameter space into orthogonal *directions* v_i
- ▶ Σ : scales each direction by σ_i
- ▶ U : rotates into data space

Each singular value σ_i measures how strongly the data responds when we move along direction v_i :

$$Av_i = \sigma_i u_i$$

Why Small Singular Values Are Poison

Key idea: Singular values quantify *visibility of directions* to the data. Large $\sigma_i \Rightarrow$ well-observed direction, small $\sigma_i \Rightarrow$ poorly observed direction. Using SVD, the OLS solution decomposes as

$$x_{\text{OLS}} = \sum_{i=1}^r \frac{1}{\sigma_i} v_i (u_i^\top b)$$

Problem with OLS:

- ▶ Small $\sigma_i \Rightarrow$ large $\frac{1}{\sigma_i}$
- ▶ Noise in $u_i^\top b$ gets *amplified*
- ▶ Leads to high variance and unstable coefficients

How Ridge Fixes It (Extracting Important Features)

Ridge regression modifies the weight:

$$x_{\text{ridge}} = \sum_{i=1}^r \left(\frac{\sigma_i}{\sigma_i^2 + \lambda} \right) v_i (u_i^\top b)$$

Effect of Ridge:

- ▶ Large σ_i : behaves like OLS (important directions preserved)
- ▶ Small σ_i : contributions suppressed (unstable directions damped)

Conclusion: Ridge does not select features directly; it selects *directions* that the data can reliably identify.

How does Ridge make training stable and converge faster?

"Regularization improves *conditioning*, conditioning controls *convergence*, and convergence determines *stability*."

If I slightly change the input, how much can the output change? That ratio is the **condition number**.

We move from solving $Ax = b$ to analyzing effect of perturbations Suppose we slightly perturb $b \rightarrow b + \delta b$, then we get $x \rightarrow x + \delta x$.

Then:

$$A(x + \delta x) = b + \delta b \quad \Rightarrow \quad A\delta x = \delta b.$$

So: $\delta x = A^{-1}\delta b$. Now we compare the relative changes: $\frac{\|\delta x\|}{\|x\|}$ vs $\frac{\|\delta b\|}{\|b\|}$.

This leads to:

$$\frac{\|\delta x\|}{\|x\|} \leq \|A^{-1}\| \|A\| \frac{\|\delta b\|}{\|b\|}.$$

That product $\|A^{-1}\| \|A\|$ is the **condition number** $\kappa(A)$.

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Thank You! Questions?